Cambridge Assessment International Education

Cambridge O Level

PHYSICS **5054/31**

Paper 3 Practical Test **May/June 2022**

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	Length and width to supervisor's results ± 1, to nearest mm only e.g. l = 298 mm ± 1 (mm) **and** w = 7 (mm) ;	**1**
1(b)	Correct answer using their result (e.g. 1.15×10^4 (mm^3)) ;	**1**
1(c)(i)	the thermometer is not a uniform cylinder / shape / has rounded ends owtte ;	**1**
1(c)(ii)	the bulb / narrow part is small compared to the rest of the thermometer ;	**1**
1(d)	temperature must be constant **or** volume of liquid varies with temperature **or** difficult to determine thickness of glass wall **or** the bulb and stem have different thicknesses ;	**1**

Question	Answer	Marks
2(a)	supervisor's value ± 2 cm ;	**1**
2(b)	*Check rule is vertical*: metre rule clamped in stand and aligned vertically with **either** correct use of set-square described **or** check the scale of the rule and the pendulum thread are parallel by measuring in two places (well apart) along the thread of the pendulum ;	**1**
	Check bob is at correct height: bottom of bob aligned with scale reading of value given in **(a)** + 10.0 cm (i.e. h_1 + 10.0 cm);	**1**
2(c)	at least 2 times averaged, divided by the number of oscillations **and** to an accuracy of 1.8 s ± 0.2 s;	**1**
2(d)	yes, **AND** correct description of one complete oscillation and the vertical distance travelled during a single oscillation (allow a correct description in terms of movement from the release point or from the resting / lowest point) and therefore the vertical speed is the vertical distance travelled in one oscillation (40) divided by the time(t) for one oscillation / or 10 divided by the time for a quarter of an oscillation, owtte ;	**1**

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Question	Answer	Marks
3(a)(i)	reading to the nearest half scale division, to at least 1d.p and in range 3.5–4.5 V ;	**1**
3(a)(ii)	Distance <30 cm;	**1**
3(a)(iii)	reading of V less than **(ai) and** difference between voltages calculated correctly ;	**1**
3(b)(i)	move/rotate the LED at **a constant distance** from the LDR and if the claim is correct, orientation in the *x* direction should produce **greater** change in V (compared to direction *y*) ORA ;	**1**
3(b)(ii)	yes or no with correct justification, linking brightness,orientation and voltage (assume at a fixed distance) based on further measurements of V seen ;	**1**

Question	Answer	Marks
4(a)	(move lens until) the image is larger than the object ; and **any two** from: adjust the height (by adding masses) until the lens height= height of centre of screen and / or height of centre of cross wire; focus the image by slowly moving the lens / back and forth ensure object, lens and screen are perpendicular to the metre rule / are all parallel to each other (by turning the lens) ;;	**3**
4(b)	*x* from 76 to 82 cm ;	**1**
4(c)	$d = 20 \pm 2$ mm AND $D = 87 \pm 10$ mm ;	**1**
4(d)	headings and units and first line of table; $\checkmark_H$ six pairs of readings of x; $\checkmark_R$ whole table complete and calculations of m correct; $\checkmark_{calc}$ consistent recording of data in each column to <u>2 or 3 s.f.;</u> $\checkmark_{SF}$	**4**

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Question	Answer	Marks
4(e)	axes labelled and correct orientation (magnification on y-axis, distance x on x-axis); ✓$_A$ suitable scale, plots occupying ⩾ ½ page in both directions; ✓$_s$ points plotted correctly; ✓$_p$ best fit straight line and fine single line and fine points or crosses; ✓$_L$	**4**
4(f)(i)	calculation shown **and** *f* in the range from 18.3 to 14.8 ;	**1**
4(f)(ii)	yes or no with appropriate justification based on how similar / different the values are ;	**1**

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